

Use Ratio Reasoning

Lesson 3-6

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$$2/3 = 8/12$$

Proportion

A math sentence saying two ratios are equal.

$$2/3 = 8/12 \rightarrow 2 \times 12 = 24 \text{ and } 3 \times 8 = 24 \checkmark$$

Cross-multiply

Multiplying across two ratios to check if they are equal.

$$5:8 \times 3 \rightarrow 15:24$$

Scale

To multiply or divide both parts of a ratio by the same number.

$$1/2 \text{ and } 3/6 \text{ and } 5/10 \text{ all equal the same amount — } \textit{half}$$

Equivalent

Having the same value.

$$60 \text{ miles in } 3 \text{ hours} \rightarrow 20 \text{ miles per } 1 \text{ hour}$$

Unit rate

A rate for just 1 of something. You find it by dividing.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can use ratio reasoning to solve problems with proportions and scaling.

1 Notice

Chef Academy received a huge catering order — a banquet for 120 guests!

2 Key idea

I can use ratio reasoning to solve problems with proportions and scaling.

3 Words I'll use

Proportion

Cross-multiply

Scale

Equivalent

Unit rate



Think About It

- How many times bigger is 120 compared to 8?
- What operation would help you scale both ingredients?
- What stays the same about the recipe even when the amounts change?



Watch

Watch your teacher model one example. Jot what you see.



We try

Solve the next one together as a class.



You try

Now try one on your own.

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Chef Reyes's recipe serves 8 people, but the banquet has 120 guests. How many times bigger is 120 than 8, and how does that scale factor help you adjust the tomatoes and mozzarella?

Level 1 support

Start here: Find the scale factor ($120 \div 8 = 15$), then multiply each ingredient by it.



Need a hint?

Sentence starters (optional)

120 guests is ____ times bigger than 8 because 120 divided by 8 equals ____.

120 invitados es ____ veces más que 8 porque 120 entre 8 es ____.

So I scale by ____, which keeps the recipe in ____.

Entonces escalo por ____, lo que mantiene la receta en ____.

Word bank: proportion scale equivalent unit rate ratio

Listen for: Students find the scale factor of 15 and explain that multiplying both ingredients by 15 keeps the recipe proportional (5 cups tomatoes becomes 75, 3 cups mozzarella becomes 45).

Level 2

What stays the same about the recipe even when all the amounts get bigger? Justify why scaling keeps the flavor the same.

- The ____ between tomatoes and mozzarella stays the same because ____.
- Scaling keeps the flavor because ____.

EXPLORE

You sorted ratio pairs into equivalent and not equivalent. How does ratio reasoning, like cross-multiplying, prove that 4:7 and 12:21 are equivalent?

Level 1 support

Start here: Cross-multiplying gives equal products when two ratios form a true proportion.

 **Need a hint?**

Sentence starters (optional)

I reasoned that 4:7 and 12:21 are ____ because ____.

Razoné que 4:7 y 12:21 son ____ porque ____.

Cross-multiplying gives ____ and ____, which are ____.

La multiplicación cruzada da ____ y ____, que son ____.

Word bank: proportion equivalent scale cross-multiply ratio

Listen for: Students show $4 \times 21 = 84$ and $7 \times 12 = 84$ (or that 4:7 scales by 3 to 12:21), proving the ratios are equivalent.

Level 2

Two ratios cross-multiply to UNEqual products. Explain what that tells you and give an example you tested.

- If the cross products are not equal, then ____.
- An example I tested is ____ because ____.

CONNECT

Where might ratio reasoning help you make a smart choice in real life, like shopping or cooking?

Level 1 support

Start here: Scaling recipes, comparing prices, and mixing ingredients all use ratio reasoning.



Need a hint?

Sentence starters (optional)

Ratio reasoning helps me when ____.

El razonamiento con razones me ayuda cuando ____.

I used the ratio to decide ____.

Usé la razón para decidir ____.

Word bank: **proportion** **scale** **equivalent** **unit rate** **ratio**

Listen for: Students name a real use (scaling a recipe, comparing deals, mixing paint or fuel) and explain how keeping the ratio equivalent guides the decision.

Level 2

Describe a situation where ignoring the ratio would cause a real problem, and explain how proportional reasoning fixes it.

- Ignoring the ratio would cause a problem when ____.
- Proportional reasoning fixes it by ____.

Guided Examples

Example 1

A recipe uses 4 cups of broth for every 10 servings. How many cups of broth are needed for 30 servings?

Solution: Scale factor: $30 \div 10 = 3$. Multiply broth by 3: $4 \times 3 = 12$ cups.

Answer: A. 12

Example 2

Are the ratios 6:9 and 2:3 equivalent?

Solution: Divide 6:9 by 3 to get 2:3. Since both ratios simplify to 2:3, they are equivalent.

Answer: A. Yes, both simplify to 2:3

Example 3

If 3 pounds of apples cost \$6, how much do 7 pounds cost?

Solution: Unit rate: $\$6 \div 3 = \2 per pound. For 7 pounds: $\$2 \times 7 = \14 .

Answer: A. \$14

Write About the Math

The Writing Revolution

I can explain my reasoning using the words proportion, scale, equivalent, and unit rate.

1. Kernel Sentence subject + verb

Model: Scale is to multiply or divide both parts of a ratio by the same number.

Escalar es multiplicar o dividir ambas partes de una razón por el mismo número.

Write a kernel sentence about scale. Use a subject and a verb.

Escribe una oración base sobre escalar. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Scale matters in math

Escalar importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Scale matters in math because ____.

Escalar importa en matemáticas porque ____.

but
pero

Scale matters in math, but ____.

Escalar importa en matemáticas, pero ____.

so
entonces

Scale matters in math, so ____.

Escalar importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about scale.
Di un hecho verdadero sobre scale.

Scale ____.

Question
Pregunta

Ask a question about scale.
Haz una pregunta sobre scale.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about scale.
Muestra entusiasmo sobre scale.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with scale.
Dile a un compañero qué hacer con scale.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning use a sentence starter

120 guests is ____ **times bigger than 8 because 120 divided by 8 equals** ____.

120 invitados es ____ *veces más que 8 porque 120 entre 8 es* ____.

So I scale by ____ , **which keeps the recipe in** ____.

Entonces escalo por ____ , *lo que mantiene la receta en* ____.

I reasoned that 4:7 and 12:21 are ____ **because** ____.

Razoné que 4:7 y 12:21 son ____ *porque* ____.

Try It

Solve on your own. Check the answer key when you are done.

1. A chef uses the proportion $\frac{3}{7} = \frac{x}{28}$ to scale a recipe. What is x ?

- A. 12
- B. 14
- C. 9
- D. 21

Show your work:

2. A recipe serves 6 people and uses 4 cups of rice. You need to serve 15 people. Show two different methods to find how much rice you need: (1) using a scale factor and (2) using a unit rate.

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Find Priya's Mistake — find the error, then write the correct reasoning.

Show your work:

Reflect — Exit Ticket

A recipe calls for 6 cups of flour for every 15 cookies. How many cups of flour are needed to make 45 cookies?

A. 18

B. 12

C. 24

D. 21

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 12 — *Scale factor: $28 \div 7 = 4$. Multiply: $3 \times 4 = 12$. So $x = 12$.*
2. **Try It 2:** Method 1 (scale factor): $15 \div 6 = 2.5$. Multiply rice by 2.5: $4 \times 2.5 = 10$ cups. Method 2 (unit rate): $4 \div 6 = 2/3$ cup per person. For 15 people: $2/3 \times 15 = 10$ cups. Both methods give 10 cups of rice.
3. **Exit Ticket:** A. 18 — *Scale factor: $45 \div 15 = 3$. Multiply flour by 3: $6 \times 3 = 18$ cups of flour.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Scale is to multiply or divide both parts of a ratio by the same number.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).